# Word in noise - Insight46 phase 1

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## Document details

| | |
|---|---|
| **Categories of variables** | Imaging outcome measure – normal appering white mtter (NAWM) diffusion tensor imaging (DTI) parameters indexing microstructural integrity including:<br><br>1. fractional anisotropy (FA)<br><br>2. mean diffusivity (MD)<br><br>3. neurite orientation dispersion and density imaging (NODDI) derived parameters of neurite density index (NDI)<br><br>4. neurite orientation dispersion and density imaging (NODDI) derived parameters of orientation dispersion index (ODI)<br><br>Alterations of NAWM integrity can be assessed by diffusion tensor imaging (DTI) which enables sensitive estimation of white matter tissue microstructure environments using water diffusion properties. Parameters typically include fractional anisotropy (FA), a measure of directionality, and mean diffusivity (MD), a measure of magnitude of diffusion. Additional multicompartmental modelling techniques have been developed to address limitations in these conventional measures. For example, neurite orientation dispersion and density imaging (NODDI) estimates microstructural complexity parameters of neurite density index (NDI), a measure of fraction of axons and dendrites; and orientation dispersion index (ODI), a measure of variability of neurite orientation (5). Lower values for FA and NDI and higher values for MD and ODI are thought to represent worse microstructural integrity. |
| **Summary of work undertaken** | Cleaning±<br><br>**Description/Rationale:**<br><br>DTI cleaning pipeline requires:<br><br>QC'd DTI scans: shell b700 (for FA, MD) and shell b2000 (for NODDI ODI FICVF)<br><br>To generate NAWM metrics requires:<br><br>QC'd DTI scans<br><br>QC'd T1, T2 and FLAIR<br><br>BaMoS final WMH masks<br><br>FA, MD and NODDI masks<br><br>**Values:**<br><br>Mean NAWM values for FA, MD, ODI and FICVF were generated. |
| **Names of source variables** | is, the, only, variable, of, interest, that, is, extracted, from, these, source, files, for, the, final, datafile |
| **Output variables** | qc_nawm_all_i46p1, qc_nawm_all_noddi_i46p1, qc_nawmreason_i46p1, nawmz_fa_f_i46p1, nawmz_md_f_i46p1, nawmz_ficvf_f_i46p1, nawmroa_fa_f_i46p1, nawmroa_md_f_i46p1, nawmz_odi_f_i46p1, nawmroa_ficvf_f_i46p1, nawmroa_odi_f_i46p1 |
| **Papers using these variables** | In progress |

## Background

Alterations of NAWM integrity can be assessed by diffusion tensor imaging (DTI) which enables sensitive estimation of white matter tissue microstructure environments using water diffusion properties. Parameters typically include fractional anisotropy (FA), a measure of directionality, and mean diffusivity (MD), a measure of magnitude of diffusion. Additional multicompartmental modelling techniques have been developed to address limitations in these conventional measures. For example, neurite orientation dispersion and density imaging (NODDI) estimates microstructural complexity parameters of neurite density index (NDI), a measure of fraction of axons and dendrites; and orientation dispersion index (ODI), a measure of variability of neurite orientation (5). Lower values for FA and NDI and higher values for MD and ODI are thought to represent worse microstructural integrity.

| **DTI NAWM: reason failed** | Freq. | Percent | Cum. |
|---|---|---|---|
| pass | 411 | 81.87 | 81.87 |
| T1 fail | 3 | 0.6 | 82.47 |
| FLAIR fail | 3 | 0.6 | 83.07 |
| BAMOS segmentation fail | 1 | 0.2 | 83.27 |
| cortical stroke mis-segmented | 5 | 1 | 84.26 |
| demyelination mis-segmentated | 3 | 0.6 | 84.86 |
| vascular anommaly mis-segmented | 1 | 0.2 | 85.06 |
| no scan | 31 | 6.18 | 91.24 |
| short scan protocol (no dti) | 17 | 3.39 | 94.62 |
| QC DTI B700 fail | 14 | 2.79 | 97.41 |
| QC DTI B2000/noddi fail | 13 | 2.59 | 100 |

| | Freq. | Percent | Cum. |
|---|---|---|---|
| **in46_w1_qc_nawm_all** | | | |
| 0 = didn't pass | 78 | 15.54 | 15.54 |
| 1 = pass | 424 | 84.46 | 100 |

| | Freq. | Percent | Cum. |
|---|---|---|---|
| **in46_w1_qc_nawm_all_noddi** | | | |
| 0 = didn't pass | 91 | 18.13 | 18.13 |
| 1 = pass | 411 | 81.87 | 100 |

## Variables in this document

### 11 high-confidence matches

| Variable | Label | How it was found |
|---|---|---|
| **Sub-Studies [57] › Insight46 [699] › Brain MRI [6033] › Normal appearing white matter (NAWM) [60333]** | | |
| nawmroa_fa_f_i46p1 | QC'd normal appearing white matter (nawm) mean fractional anisotropy (fa) - raw value - Cross-sectional - Insight46 phase 1 | topsheet field |
| nawmroa_ficvf_f_i46p1 | QC'd normal appearing white matter (nawm) mean intracellular volume fraction (ficvf) raw value - Cross-sectional - Insight46 phase 1 | topsheet field |
| nawmroa_md_f_i46p1 | QC'd normal appearing white matter (nawm) mean diffusivity (md) - raw value - Cross-sectional - Insight46 phase 1 | topsheet field |

| Variable | Label | How it was found |
|---|---|---|
| nawmroa_odi_f_i46p1 | QC'd normal appearing white matter (nawm) mean orientation dispersion index (odi) - raw value - Cross-sectional - Insight46 phase 1 | topsheet field |
| nawmz_fa_f_i46p1 | QC'd normal appearing white matter (nawm) mean fractional anisotropy (fa) - standardised to control sample - Cross-sectional - Insight46 phase 1 | topsheet field |
| nawmz_ficvf_f_i46p1 | QC'd normal appearing white matter (nawm) mean intracellular volume fraction (ficvf) - standardised to control sample - Cross-sectional - Insight46 phase 1 | topsheet field |
| nawmz_md_f_i46p1 | QC'd normal appearing white matter (nawm) mean diffusivity (md) - standardised to control sample - Cross-sectional - Insight46 phase 1 | topsheet field |
| nawmz_odi_f_i46p1 | QC'd normal appearing white matter (nawm) mean orientation dispersion index (ODI) - standardised to control sample - Cross-sectional - Insight46 phase 1 | topsheet field |
| qc_nawm_all_i46p1 | Diffusion tensor imaging (DTI) all reasons: 0 (don't use) 1 (use) - Cross-sectional - Insight46 phase 1 | topsheet field |
| qc_nawm_all_noddi_i46p1 | Diffusion tensor imaging (DTI) neurite orientation dispersion and density imaging (NODDI) all reasons: 0 (don't use) 1 (use) - Cross-sectional - Insight46 phase 1 | topsheet field |
| qc_nawmreason_i46p1 | Diffusion tensor imaging (DTI) normal appearing white matter (NAWM): reason failed - Cross-sectional - Insight46 phase 1 | topsheet field |

## 1 medium-confidence match

| Variable | Label | How it was found |
|---|---|---|
| **Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Heart and circulatory conditions [51014]** | | |
| stroke | Have you had a stroke in the last 10 years - at age 53 years | prose scan |